



TFG – Resolution Studies

Residual Computation

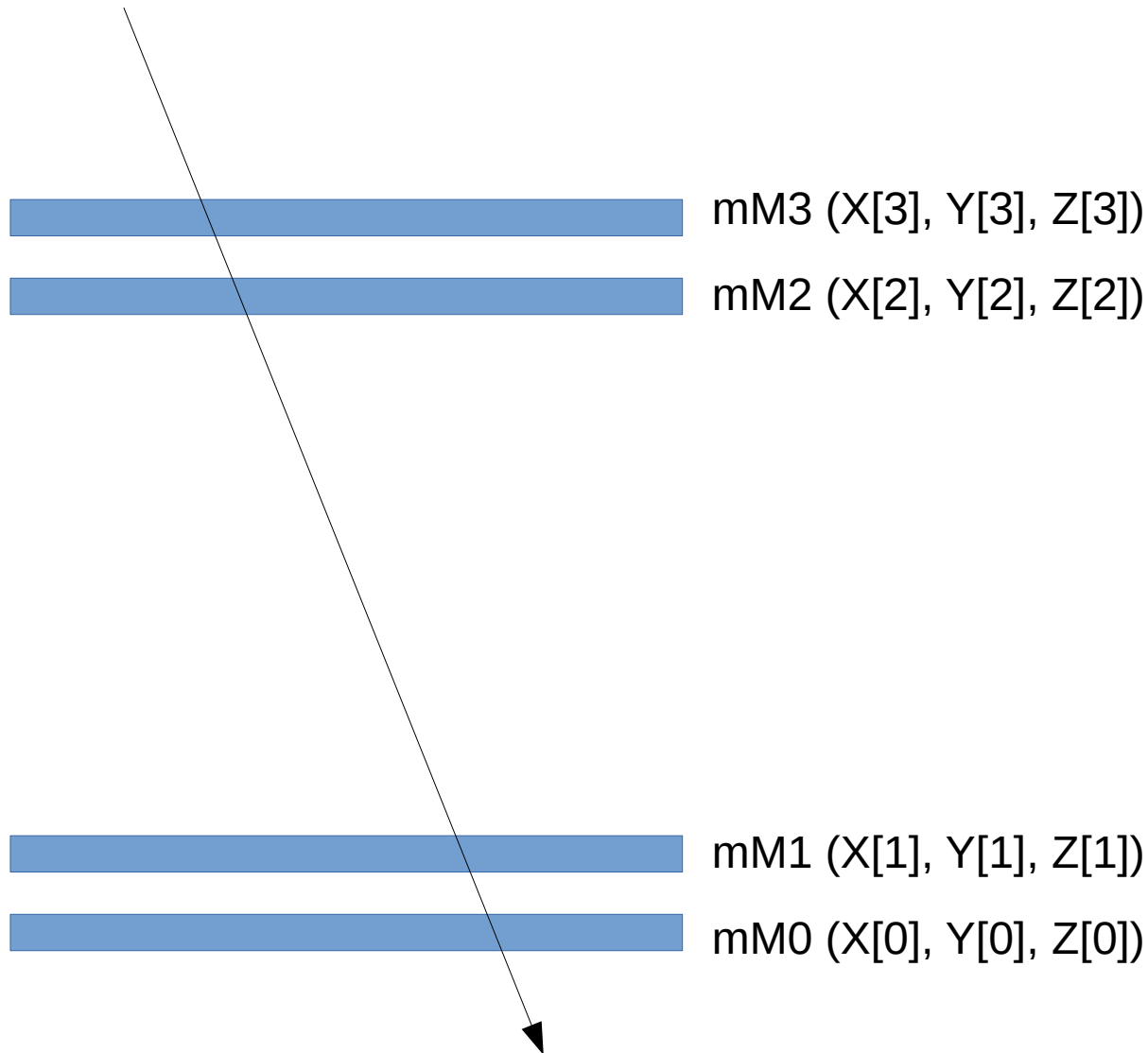
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As we have already mention, an event in our muon telescope is a set of points containing the position of the muon interaction in each Micromegas detector:

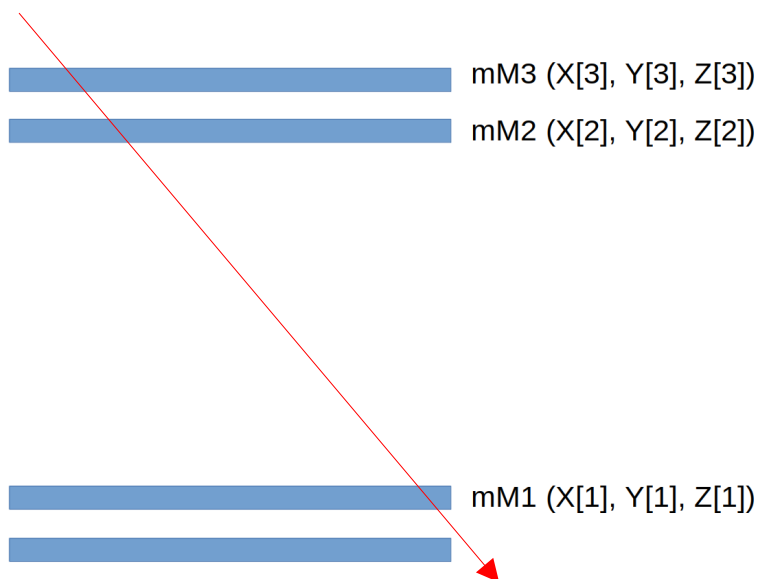
→ For experimental data these points are computed from the readout signal (signal processing)

→ For simulations each point is the real interaction point

From this set of points: How do we do the muon track fit and we compute the corresponding residual?

Note: This is the historical way the Irfu's muography group used. We can take as baseline but in a further phase we can consider to change this method

Case 1: Three triggered detectors (out of four)



0.- Only possible scenarios are mMs 1, 2 and 3 triggered or mMs 0, 1 and 2. Both are equivalent.

Note: We will discuss about some particularities of experimental data (keep in mind)

1.- For each event we can compute ResX taking the pairs (X, Z) and ResY taking the pairs (Y, Z). Again, both are equivalent

2.- For ResX, we compute the line passing through (X_1, Z_1) and (X_3, Z_3). It has a perfect solution

3.- From this line we compute the corresponding point at the middle detector (X'_2, Z_2)

4.- We estimate ResX as the distance between the interpolated point and the real point

$$\text{ResX} = \text{abs}(X_2 - X'_2)$$

5.- We do the analog to compute ResY

From this set of points: How do we do the muon track fit and we compute the correspondig residual?

Note: This is the historical way the Irfu's muography group used. We can take as baseline but in a further phase we can consider to change this method

Case 1: All Four triggered detectors

mM3 (X[3], Y[3], Z[3])
mM2 (X[2], Y[2], Z[2])

mM1 (X[1], Y[1], Z[1])
mM0 (X[0], Y[0], Z[0])

Possible 3 detectors combinations

0	1	2
0	1	3
0	2	3
1	2	3

0.- We can convert this case in a 3 detector case following the table on the left. There are 4 possible combinations

1.- For each combination of detectors we compute ResX and ResY as in the 3 case detectors (previous slide)

2.- we obtain **4 values for ResX and 4 values for ResY.**

3.- We take the **smallest value of each of them** to determine ResX and ResY

Note: Is possible that the smallest ResX comes from a detector combination different than for the smallest ResY. It is not a problem